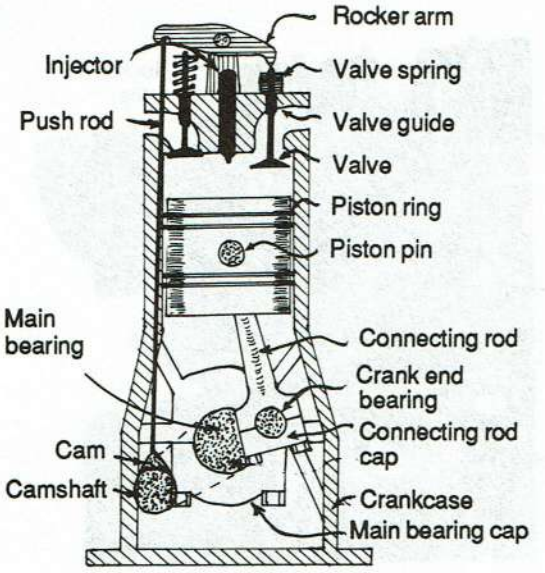


Figure 1-8. *Principal components in a diesel engine.*

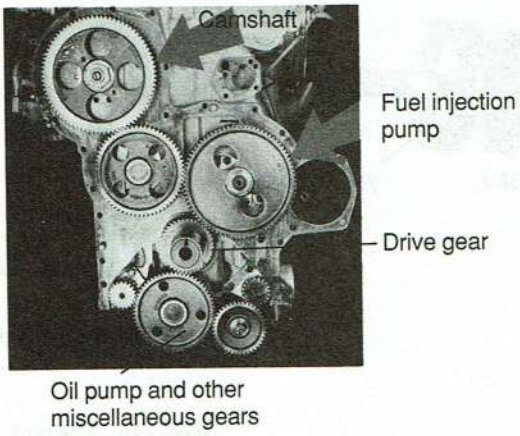


Figure 1-9. *Engine timing gears. (Courtesy Caterpillar Tractor Co.)*

with an intermediate gear, a belt, or a chain so that they rotate together (see Figure 1-9).

On a 4-cycle engine, the inlet and exhaust valves are opened and closed on every other stroke of the engine. As a consequence, the gear on the camshaft is twice the size of that on the crankshaft, causing the former to rotate at half the speed of the latter and the valves to open and close on every other engine revolution.

Detroit Diesel 2-cycle engines have only exhaust valves. The exhaust valves open on every downward stroke of the piston; therefore, the camshaft gear is the same size as that on the crankshaft. The two shafts rotate at the same speed, and the valves open and close on every engine revolution.

By setting the camshaft and crankshaft gears in different relationships with one another, the valves can be made to open and close at any point of piston travel. Timing engine valves consists of placing the gears in the precise relationship needed for overall maximum engine performance.

Fuel injection timing is set the same way. A gear fitted to the end of the fuel-injection pump drive shaft is driven by the crankshaft via an intermediate gear, belt, or chain. Altering the relationship of the gears allows the fuel injection to be set at any point of piston travel. Since injection occurs on every other revolution on a 4-cycle engine, the pump drive gear is twice the size of the crankshaft gear, causing the pump to rotate at half the engine's speed. On 2-cycle engines, the gears are the same size, producing injection at every engine revolution.

Cylinders and Other Parts

The central frame to which the rest of an engine is assembled is known as the *block*, or *crankcase* (see Figure 1-10). On all except air-cooled engines, which are rarely found on boats, this is a casting containing numerous passages for air, cooling water, oil, and other engine parts, such as the crankshaft and camshaft.

There are two types of cylinders, or *liners*, available for diesel engines—dry liners and wet liners. In a dry-liner diesel the cylinders and block are in a continuous metal-to-metal contact. In a wet-liner arrangement the block comes into contact with the cylinders only at their tops and bottoms, and engine cooling water circulates directly around the liners (see Figure 1-11). Wet liners have the advantage of being relatively easy to replace in the field when a major engine overhaul is necessary, whereas the whole block must generally be taken to a machine shop to change dry liners.

Pistons are fitted with a number of *piston rings*, set in grooves cut into the outside circumference of the